

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

08

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

A cylinder has a diameter of 8 inches and a height of 12 inches. What is the volume, in cubic inches, of the cylinder?

- A. 16π
- B. 96π
- C. 192π
- D. 768π

Q2

GRID-IN

GEOMETRY AND TRIGONOMETRY

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The table gives the perimeters of similar triangles TUV and XYZ , where TU corresponds to XY . The length of TU is 18.

	Perimeter
Triangle TUV	37
Triangle XYZ	333

What is the length of XY ?

- A. 2
- B. 18
- C. 55
- D. 162

A triangle has a base length of 56 centimeters and a height of 112 centimeters. What is the area, in square centimeters, of the triangle?

- A. 168
- B. 1,568
- C. 3,136
- D. 6,272

A manufacturing company produces two sizes of cylindrical containers that each have a height of 50 centimeters. The radius of container A is 16 centimeters, and the radius of container B is 25% longer than the radius of container A. What is the volume, in cubic centimeters, of container B?

- A. $16,000\pi$
- B. $20,000\pi$
- C. $25,000\pi$
- D. $31,250\pi$

Square A has side lengths that are 246 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

- A. 60,516
- B. 492
- C. 246
- D. 123

Q7

GRID-IN

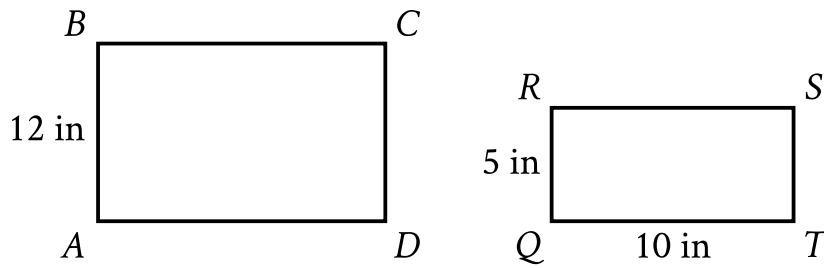
GEOMETRY AND TRIGONOMETRY

What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



Note: Figure not drawn to scale.

- Rectangle upper A upper B upper C upper D is labeled as follows:
 - The length of side upper A upper B is 12 inches.
- Rectangle upper Q upper R upper S upper T is labeled as follows:
 - The length of side upper Q upper R is 5 inches.
 - The length of side upper Q upper T is 10 inches.
- A note indicates the figure is not drawn to scale.

Rectangles $ABCD$ and $QRST$ shown are similar, where A , B , C , and D correspond to Q , R , S , and T , respectively. What is the length, in inches in, of $\overline{AD}$?

- A. 60
- B. 24
- C. 17
- D. 10

A right circular cylinder has a height of 8 meters m and a base with a radius of 12 m. What is the volume, in m^3 , of the cylinder?

- A. 8π
- B. 20π
- C. 768π
- D. $1,152\pi$

A right circular cylinder has a volume of 432 cubic centimeters. The area of the base of the cylinder is 24 square centimeters. What is the height, in centimeters, of the cylinder?

- A. 18
- B. 24
- C. 216
- D. 10,368